

Summary

Materials scientist with a Ph.D. in Chemistry and postdoctoral experience at Argonne National Laboratory. Experienced in polymer and electrochemical materials, with strengths in materials synthesis, characterization, and structure–property analysis. Author of 13 peer-reviewed publications, including three first-author papers.

Skills

- **Characterization:** multinuclear NMR (^1H , ^{13}C , ^{15}N , ^{19}F , ^{31}P ; variable-temperature and 2D NMR); FTIR; Raman; UV–Vis; fluorescence spectroscopy; Synchrotron SAXS/GISAXS at the Advanced Photon Source; DLS; GPC; HPLC; GC-MS; GC-FID
- **Electrochemical Methods:** Cyclic voltammetry (CV); electrochemical impedance spectroscopy (EIS)
- **Materials Synthesis:** Polymer membrane synthesis and fabrication; thin-film preparation; transition-metal complex synthesis; inert-atmosphere synthesis
- **Data Analysis and Research Tools:** Mnova; Origin; Python; scientific data analysis and visualization
- **Laboratory Operations:** SOP development; EHS protocol implementation; laboratory setup; GPC instrument specification evaluation and vendor coordination

Professional Experience

Postdoctoral Appointee – Argonne National Laboratory Jan 2025 – Present

- Developed reproducible polymer synthesis protocols by controlling solvent and reagent purity and monitoring polymerization by GPC, enabling consistent molecular weight and dispersity.
- Evaluated polymer molecular weight, dispersity, and ionic conductivity to qualify membrane samples for delivery to industrial collaborators.
- Performed SAXS/GISAXS measurements and data analysis at the Advanced Photon Source for polymer morphology characterization.
- Established SOPs and participated in safety reviews for a new polymer membrane synthesis laboratory; evaluated GPC system specifications and coordinated with vendors.
- Inventor on a pending patent for fuel-cell membrane technology.

Organic Chemist (Intern) – KLA May – Aug 2024

- Synthesized rhodamine-based photoswitchable dyes for inspection technology development and characterized optical properties using UV–Vis and fluorescence spectroscopy.

Graduate Teaching/Research Assistant – Michigan State University Sep 2018 – Dec 2024

- Investigated Ru-catalyzed ammonia oxidation using variable-temperature NMR, isotope labeling, and Raman spectroscopy.
- Identified a previously uncharacterized dinitrogen-bridged intermediate, leading to a first-author publication in the *Journal of the American Chemical Society*.
- Led multinuclear NMR characterization and mechanistic analysis in a multidisciplinary electrochemical research collaboration, resulting in a first-author publication in *Inorganic Chemistry*.
- Trained junior researchers and served as a teaching assistant for General and Organic Chemistry.

Education

Ph.D. in Chemistry, Michigan State University, USA	2018 – 2024
M.S. in Chemistry, National Sun Yat-Sen University, Taiwan	2012 – 2014
B.S. in Chemistry, Chung Yuan Christian University, Taiwan	2008 – 2012

Research Highlights

- 13 peer-reviewed publications, including 3 first-author papers in *JACS*, *Inorganic Chemistry*, and *ACS Applied Materials & Interfaces* (full list on [Google Scholar](#))
- 1 patent application pending
- Oral presentations at ACS Fall **2026**, 249th ECS Meeting **2026**, and ACS Fall **2023** national meetings